

Value Function Approximation — Rollout Layer

improves drop decisions by replacing greedy selection
with forward-looking Monte-Carlo rollout evaluation

Myopic

$$\text{score} = \frac{1}{d_{\text{cur}, i}}$$

$$\text{drop} = \frac{1}{\Delta t_i}$$

*greedy, no
economic weighting*

Myopic+

$$\text{score} = \frac{p_i}{d_{\text{cur}, i}}$$

$$\text{drop} = \frac{p_i}{\Delta t_i}$$

*+ power-weighted
urgency*

CFA

$$\text{score} = \frac{\tilde{C}_i + s}{d_{\text{cur}, i}}$$

$$\text{drop} = \tilde{C}_i - \frac{c_L}{60} \Delta t_i$$

*+ learned station
value $\tilde{C}$*

Dynamic Balance

$$\text{score} = \frac{\tilde{C}_i + s}{d_{\text{cur}, i}^{2\delta}}$$

$$\text{drop} = \tilde{C}_i - \frac{2\delta c_L}{60} \Delta t_i$$

*+ state-dependent
balance δ*

Shared Zone Selection Layer

provides each team's daily station pool via K-Means zone clustering and zone scoring